

Dengue Serotype Research: Reducing Clustering Variability

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(1) Creating Consensus Maps

01 Produce Co-Association Matrices

For each clustering run, produce a matrix of co-association containing **0** if regions are not in the same cluster, and **1** if they are in the same cluster.

02 Create Mean Co-Association Matrix

Scrape each Max-P run's "**Best Objective Value**" from Terminal using file handling. Compute a weighted (inverse **softmax**) mean of the n co-association matrices to create a "mean co-association matrix."

03 Cluster the Weighted Matrix

Feed the mean co-association matrix into **clustering algorithms** (Hierarchical, Spectral) and also test Louvain Community Detection.

```
maxp_terminal_log
total_within_region_distance after SA:
1725.791610539139
total_within_region_distance after SA:
2429.9227587520863
total_within_region_distance after SA:
1748.0005022773797
total_within_region_distance after SA:
1896.1104473105108
total_within_region_distance after SA:
2400.877727250709
total_within_region_distance after SA:
2408.3228660809728
total_within_region_distance after SA:
2387.9636135864916
total_within_region_distance after SA:
2432.0114966293077
best objective value:
1691.7084022378672
Starting clustering run 3 of 30
max_p: 29
number of good partitions: 1
0
total_within_region_distance after SA:
1697.8418161074947
total_within_region_distance after SA:
1634.5722167457056
total_within_region_distance after SA:
1725.1071832504408
total_within_region_distance after SA:
1689.8210299745904
total_within_region_distance after SA:
1718.4084780392257
total_within_region_distance after SA:
1703.4515552037797
total_within_region_distance after SA:
1719.3074291620449
total_within_region_distance after SA:
1715.0868207905728
total_within_region_distance after SA:
1682.925552365197
total_within_region_distance after SA:
1713.265508185267
total_within_region_distance after SA:
1696.587636138401
total_within_region_distance after SA:
1706.7869135773433
total_within_region_distance after SA:
```



best_obj_vals

best_objective_value
1694.086538247950
1691.7084022378700
1634.5722167457100
1700.2625534958800
1727.0482339208900
1666.1309726658700
1697.4177293014900
1719.5249359556400
1680.9312520513400
1732.616303687060
1688.7386043286500
1662.462061617750
1680.0642304923700
1593.964933722200
1693.8196474213900
1650.3949198381200
1710.3503326861200
1702.4727760819800
1725.5068351395000
1744.39188465120
1712.1982086470400
1727.1306399817700
1671.7494040199500
1702.678135643440
1679.2362094675000
1690.6620212488400
1734.2017949441300
1687.513867004780
1785.988257729200
1686.2051816203000

(1.3) Creating Consensus Maps - Visualization



hierarchical



spectral



louvain

(2) Stabilizing Consensus Maps Across Runs

```
SPATIAL_AGGREGATION="rgint"
```

```
SA_VALUES=(10 30)
```

```
N_VALUES=(200 500)
```

```
COVARIATE_SETS=("metro")
```

```
RUN_ID=72
```

```
REPS=(a b c d e)
```

```
# Log directory
```

```
TIMESTAMP=$(date +"%Y%m%d_%H%M%S")
```

```
LOG_DIR="./.log/${RUN_ID}_${SPATIAL_AGGREGATION}_${TIMESTAMP}"
```

```
mkdir -p "$LOG_DIR"
```

```
source /opt/anaconda3/etc/profile.d/conda.sh # change to where  
conda activate DENV-SEROTYPE-IMPUTATION
```

```
for N in $(N_VALUES); do  
  for SA in $(SA_VALUES); do  
    for REPS in $(REPS); do  
      for COVARIATE_SETS in $(COVARIATE_SETS); do  
        # Run the script  
        python3 run_stabilize.py  
        # Save the results  
        cp output/* "$LOG_DIR/"  
      done  
    done  
  done  
done
```

Tested different combinations of:

- **Simulated annealing steps** (10, 20, 30, 50)
- **Runs** (100, 200, 500, 1000)
- **Covariate sets** (none, two, three, five)
- **mean()** vs **median()** to compute threshold
- **Rook** vs **Queen** adjacency

Ran each combination 4-5 times
(RUN_IDa-e), checked for stability

(2) Stabilizing Consensus Maps Across Runs

Adjusted Rand Index (ARI)

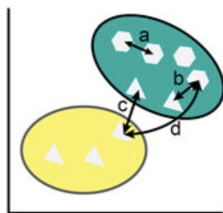
- Evaluates how similar two clusterings are after **adjusting for random chance**
- Values range from **-1 (total disagreement)** to **+1 (perfect agreement)**; 0 indicates random
- Counts how many “agreements” in region placement (into clusters) there are between clusterings A & B
- Subtracts the expected score from a random assignment

Variance of Information (VI)

- Distance metric that quantifies **how much info is lost/gained** when transitioning between two clusterings
- Values start at 0 (identical) → unbounded upper limit
- $VI(X,Y) = H(X) + H(Y) - 2I(X,Y)$
 - H is entropy (spread of cluster sizes)
 - I is mutual information (shared structure)
- # of clusters, balance of cluster sizes → entropy

ARI = pairwise consistency; VI = structural similarity

Adjusted Rand Index (ARI)



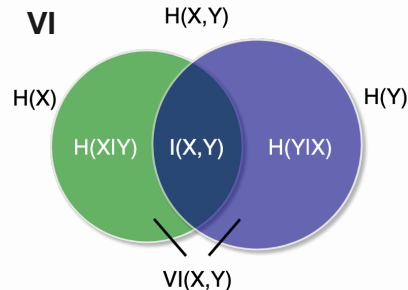
a, d: pairs of samples that agree with the ground truth.

b, c: pairs of samples that disagree with the ground truth.

$$RI = \frac{a+d}{a+b+c+d} \quad C_2^N \text{ (all possible sample pairs)}$$

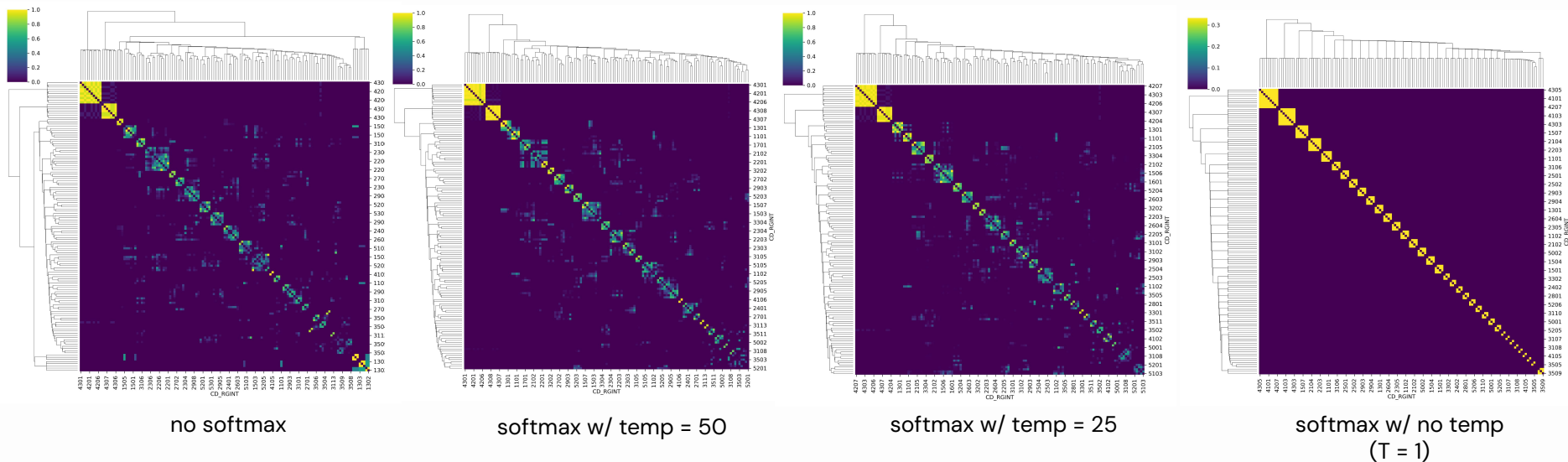
$$ARI = \frac{RI - E(RI)}{1 - E(RI)} \quad E(RI): \text{expected RI}$$

VI



(2) Stabilizing Consensus Maps Across Runs

The Effect of Softmax Temperature on Clustermaps

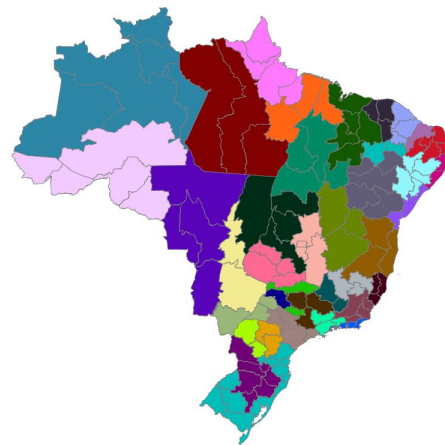
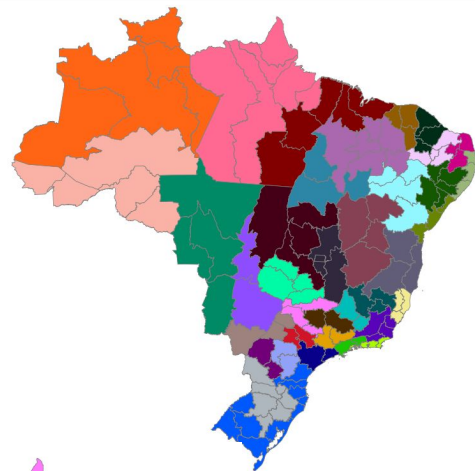


Temperature	SA Steps	# of Runs	Mean ARI	Mean VI
1	10	200	0.6580	0.8504
1	10	500	0.6355	0.9362
1	30	200	0.6768	0.8078
1	30	500	0.6839	0.8075
25	10	200	0.7660	0.5485
25	10	500	0.8310	0.3516
25	30	200	0.7995	0.4611
25	30	500	0.8189	0.4103
50	10	200	0.8194	0.3940
<u>50</u>	<u>10</u>	<u>500</u>	<u>0.8802</u>	<u>0.2667</u>
50	30	200	0.8529	0.3293
50	30	500	0.8345	0.3535

What if we remove softmax entirely?

Softmax	SA Steps	# of Runs	Mean ARI	Mean VI
No	10	200	0.7967	0.4477
No	10	500	0.8584	0.3017
No	30	200	0.8236	0.3741
No	30	500	0.8325	0.3407
Yes, T = 50	10	200	0.8194	0.3940
<u>Yes, T = 50</u>	<u>10</u>	<u>500</u>	<u>0.8802</u>	<u>0.2667</u>
Yes, T = 50	30	200	0.8529	0.3293
Yes, T = 50	30	500	0.8345	0.3535

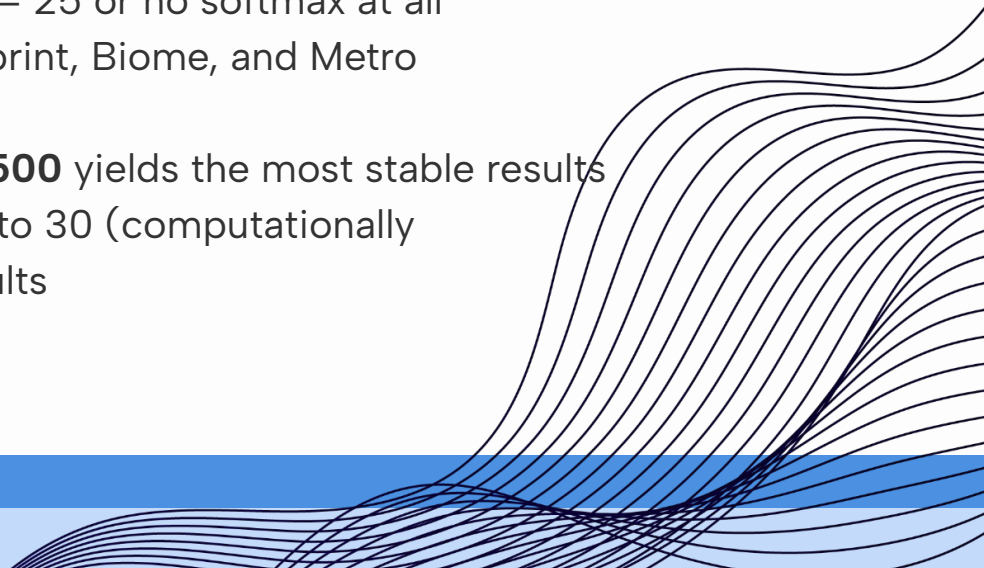
What does this stability actually look like? (Softmax with $T=50$, $SA=10$, $N=500$)



*Slight variance
exists (as expected)
– but mostly stable!*

Takeaways

- **Softmax weighting** improves agreement across runs by emphasizing higher-quality regionalizations
- **But, temperature tuning matters:**
 - $T = 1 \rightarrow$ lower ARI and high VI
 - **$T = 50$** $\rightarrow$ high ARI, low VI (strongest stability)
 - Generally slightly better than $T = 25$ or no softmax at all
- Using IndexP, Compactness, Human Footprint, Biome, and Metro covariates:
 - **$T = 50$, SA Steps = 10, # of Runs = 500** yields the most stable results
 - We don't need to increase SA Steps to 30 (computationally expensive) for strong and stable results



Thank You!

